PROBLEM 13

Quadratures & Trajectory

Conservative Force and Potential IV

Original statement · Problems.pdf, p. 9

A particle of mass m moves along the Ox axis under the potential:

$$V(x) = mgx\left[\left(\frac{x}{a}\right)^2 - 3\right]$$

Compute the evolution of the particle as a function of time, when it is released from $x_0 = a$ with an initial velocity $v_0 = \sqrt{8ga}$ directed along the positive Ox axis.

(Tip: $\int \frac{dx}{(x+1)\sqrt{2-x}} = -\frac{2}{\sqrt{3}}\operatorname{arctanh}\left(\sqrt{\frac{2-x}{3}}\right)$*).*

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A particle of mass m moves along the Ox axis subject to the cubic potential energy function:

$$V(x) = mgx\left[\left(\frac{x}{a}\right)^2 - 3\right] = mg\left(\frac{x^3}{a^2} - 3x\right)$$

- **Initial Conditions:** Released at $t = 0$ from position $x_0 = a$ with initial velocity:

$$x(0) = a, \qquad v_0 = \dot{x}(0) = +\sqrt{8ga}$$

- **Provided Integration Formula:**

$$\int \frac{dy}{(y+1)\sqrt{2-y}} = -\frac{2}{\sqrt{3}} \operatorname{arctanh}\left(\sqrt{\frac{2-y}{3}}\right) + C$$

- **Parameters:** $m > 0$ [kg], $g > 0$ [m/s^2], $a > 0$ [m].

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- **Dimensionless Coordinate:** Let $y(t) = \frac{x(t)}{a} \implies x(t) = ay(t), \dot{x}(t) = a\dot{y}(t)$.
- **Potential Energy in Dimensionless Form:**

$$V(y) = mga(y^3 - 3y)$$

- **Extrema of $V(y)$:**

$$\frac{dV}{dy} = mga(3y^2 - 3) = 3mga(y^2 - 1)$$

Setting $\frac{dV}{dy} = 0$:

- At $y = +1$ ($x = a$): $V(1) = mga(1 - 3) = -2mga$.

$\frac{d^2V}{dy^2}(1) = 6mga(1) = +6mga > 0 \implies$ **Local Minimum** (Stable equilibrium).

- At $y = -1$ ($x = -a$): $V(-1) = mga(-1 + 3) = +2mga$.

$\frac{d^2V}{dy^2}(-1) = -6mga < 0 \implies$ **Local Maximum** (Unstable equilibrium barrier).

- **Total Mechanical Energy E:**

Pedagogical Justification: Evaluating the initial energy reveals whether the particle is trapped in the well or has enough kinetic energy to reach other critical points.

$$E = T(0) + V(x_0) = \frac{1}{2}mv_0^2 + V(a)$$

$$E = \frac{1}{2}m(8ga) + mga(1^3 - 3 \cdot 1) = 4mga - 2mga = +2mga$$

Critical Analytical Discovery: The total energy $E = +2mga$ is **identical** to the potential energy at the local maximum: $E = V(-a) = +2mga$.

PHASE 3

Step-by-Step Mathematical Deduction

1. Quadrature Formulation

Pedagogical Justification: Conservation of mechanical energy $T + V = E$ gives a first-order separable differential equation for $y(t)$.

$$\frac{1}{2}m\dot{x}^2 = E - V(x) = 2mga - mga(y^3 - 3y) = mga(2 + 3y - y^3)$$

Substitute $\dot{x} = a\dot{y}$:

$$\frac{1}{2}ma^2\dot{y}^2 = mga(2 + 3y - y^3) \implies \dot{y}^2 = \frac{2g}{a}(2 + 3y - y^3)$$

2. Factorization of the Cubic Polynomial $P(y) = 2 + 3y - y^3$

Rewrite in descending powers:

$$P(y) = -(y^3 - 3y - 2)$$

Test candidate roots of $y^3 - 3y - 2 = 0$:

- For $y = -1$: $(-1)^3 - 3(-1) - 2 = -1 + 3 - 2 = 0$. Thus $(y + 1)$ is a factor.

Perform polynomial long division of $y^3 - 3y - 2$ by $(y + 1)$:

$$\frac{y^3 - 3y - 2}{y + 1} = y^2 - y - 2 = (y + 1)(y - 2)$$

Therefore:

$$y^3 - 3y - 2 = (y + 1)^2(y - 2)$$

Substituting back into $P(y)$:

$$P(y) = -(y + 1)^2(y - 2) = (y + 1)^2(2 - y)$$

Hence, the differential equation simplifies to:

$$\dot{y}^2 = \frac{2g}{a}(y + 1)^2(2 - y)$$

3. Stage 1: Forward Motion to the Turning Point ($0 \le t \le t_1$)

Initially, at $t = 0$, $y(0) = 1$ and $\dot{y}(0) = \frac{v_0}{a} = +\sqrt{\frac{8g}{a}} > 0$.

In the interval $y \in [1, 2]$, we have $y + 1 > 0$ and $2 - y \ge 0$.

Taking the positive square root:

$$\dot{y} = +\sqrt{\frac{2g}{a}}\,(y+1)\sqrt{2-y}$$

Separating variables:

$$\frac{dy}{(y+1)\sqrt{2-y}} = \sqrt{\frac{2g}{a}}\,dt$$

Integrate from $y = 1$ at $t = 0$ to $y(t)$:

$$\int_1^{y(t)} \frac{dy}{(y+1)\sqrt{2-y}} = \sqrt{\frac{2g}{a}} \int_0^t dt' = \sqrt{\frac{2g}{a}}\,t$$

Apply the provided integral hint:

$$\left[-\frac{2}{\sqrt{3}} \operatorname{arctanh}\left(\sqrt{\frac{2-y}{3}} \right) \right]_1^{y(t)} = \sqrt{\frac{2g}{a}}\,t$$

Evaluate at the lower limit $y = 1$:

$$\sqrt{\frac{2-1}{3}} = \frac{1}{\sqrt{3}}$$

$$-\frac{2}{\sqrt{3}}\operatorname{arctanh}\left(\sqrt{\frac{2-y}{3}}\right)-\left(-\frac{2}{\sqrt{3}}\operatorname{arctanh}\left(\frac{1}{\sqrt{3}}\right)\right)=\sqrt{\frac{2g}{a}}\,t$$

Multiply the entire equation by $-\frac{\sqrt{3}}{2}$:

$$\operatorname{arctanh}\left(\sqrt{\frac{2-y}{3}}\right)-\operatorname{arctanh}\left(\frac{1}{\sqrt{3}}\right)=-\frac{\sqrt{3}}{2}\sqrt{\frac{2g}{a}}\,t=-\sqrt{\frac{3g}{2a}}\,t$$

$$\operatorname{arctanh}\left(\sqrt{\frac{2-y}{3}}\right)=-\sqrt{\frac{3g}{2a}}\,t+\operatorname{arctanh}\left(\frac{1}{\sqrt{3}}\right)$$

4. Turning Point at $y=2$ ($x=2a$)

The turning point occurs when $\dot{y}=0$, which corresponds to $2-y=0 \implies y=2$.

At $y=2$, $\operatorname{arctanh}(0)=0$.

The time t_1 required to reach the turning point is:

$$0=-\sqrt{\frac{3g}{2a}}\,t_1+\operatorname{arctanh}\left(\frac{1}{\sqrt{3}}\right) \implies t_1=\sqrt{\frac{2a}{3g}}\operatorname{arctanh}\left(\frac{1}{\sqrt{3}}\right)$$

5. Stage 2: Reflected Motion towards the Barrier ($t>t_1$)

At $y=2$, the force acting on the particle is:

$$F_x(2a)=-\frac{dV}{dx}(2a)=-\frac{3mg}{a^2}(4a^2-a^2)=-9mg<0$$

Because the force is strictly negative, the particle accelerates to the left, so for $t>t_1$, $\dot{y}<0$:

$$\dot{y} = -\sqrt{\frac{2g}{a}}\,(y+1)\sqrt{2-y}$$

Separating variables from the turning point $(y = 2, t = t_1)$ to $y(t)$:

$$\int_2^{y(t)} \frac{dy}{(y+1)\sqrt{2-y}} = -\sqrt{\frac{2g}{a}} \int_{t_1}^{t} dt' = -\sqrt{\frac{2g}{a}}\,(t - t_1)$$

Using the integral formula and noting that at $y = 2$, $\operatorname{arctanh}(0) = 0$:

$$-\frac{2}{\sqrt{3}} \operatorname{arctanh}\left(\sqrt{\frac{2-y}{3}}\right) - 0 = -\sqrt{\frac{2g}{a}}\,(t - t_1)$$

Multiply by $-\frac{\sqrt{3}}{2}$:

$$\operatorname{arctanh}\left(\sqrt{\frac{2-y}{3}}\right) = \sqrt{\frac{3g}{2a}}\,(t - t_1) = \sqrt{\frac{3g}{2a}}\,t - \sqrt{\frac{3g}{2a}}\,t_1$$

Since $\sqrt{\frac{3g}{2a}}\,t_1 = \operatorname{arctanh}\left(\frac{1}{\sqrt{3}}\right)$:

$$\operatorname{arctanh}\left(\sqrt{\frac{2-y}{3}}\right) = \sqrt{\frac{3g}{2a}}\,t - \operatorname{arctanh}\left(\frac{1}{\sqrt{3}}\right)$$

PHASE 4

Physical Interpretation & Dimensional Verification

- **Asymptotic Behavior as $t \to \infty$:**

As $t \to \infty$, the right-hand side approaches $+\infty$. Because $\operatorname{arctanh}(u) \to +\infty$ as $u \to 1^{-}$:

$$\sqrt{\frac{2-y}{3}} \to 1 \implies \frac{2-y}{3} \to 1 \implies 2-y \to 3 \implies y \to -1$$

Consequently, $x(t) = a y(t) \to -a$.

The particle asymptotically approaches the unstable potential barrier at $x = -a$ over an infinite duration, never crossing it.

- **Maximum / Terminal Speed Verification:**

During the return motion, the particle passes back through the potential minimum $x = a$ ($y = 1$), where its speed returns to its maximum value:

$$v = \sqrt{\frac{2}{m}[E - V(a)]} = \sqrt{\frac{2}{m}[2mga - (-2mga)]} = \sqrt{\frac{8mga}{m}} = \sqrt{8ga}$$

which corresponds to the characteristic speed referenced in Problems.pdf page 72.

- **Dimensional check:** $[\dot{y}^2] = [2g/a] = \mathrm{s}^{-2}$, the argument of $\operatorname{arctanh}$ is dimensionless, and $[t_1] = \sqrt{[a]/[g]} = \mathrm{s}$.
- **Numerical check (RK4, step 10^{-5}, $g = a = 1$):** integrating $\ddot{y} = -(3g/a)(y^2 - 1)$ from $y = 1, \dot{y} = \sqrt{8g/a}$ gives the turning point $y = 2$ at $t = 0.53765\sqrt{a/g}$, in agreement with $t_1 = \sqrt{2a/(3g)}\,\operatorname{arctanh}(1/\sqrt{3}) = 0.537646\sqrt{a/g}$, and $y(0.3) = 1.759553$ in agreement with the closed form $y = 2 - 3\tanh^2\left(\operatorname{arctanh}(1/\sqrt{3}) - \sqrt{3g/(2a)}\,t\right) = 1.759553$.

Source: Problems.pdf (Page 9)